

TrES-5b

R-1762

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_130704 · created 2026-08-02T21:05:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456477.859 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.051
Transit depth (Rp/Rs)^2	2.63 +/- 1.66 [%]
Orbital Inclination (inc)	83.12 +/- 2.49
Airmass coefficient 1 (a1)	0.875 +/- 0.041
Airmass coefficient 2 (a2)	0.11 +/- 0.41
Scatter in the residuals of the lightcurve fit is	4.65 %
Best Comparison Star	#1 - [325, 131]
Optimal Aperture	2.55
Optimal Annulus	6.26
Transit Duration (day)	0.063 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.162 $\pm$ 0.051 vs 0.142 (deviation 0.36 σ)
Duration fit vs published	0.063 d vs 0.0758 d (deviation 0.54 σ)
Mid-time O-C	-3.5 min
Depth SNR	3.0
Residual scatter	4.65 %

Light curve

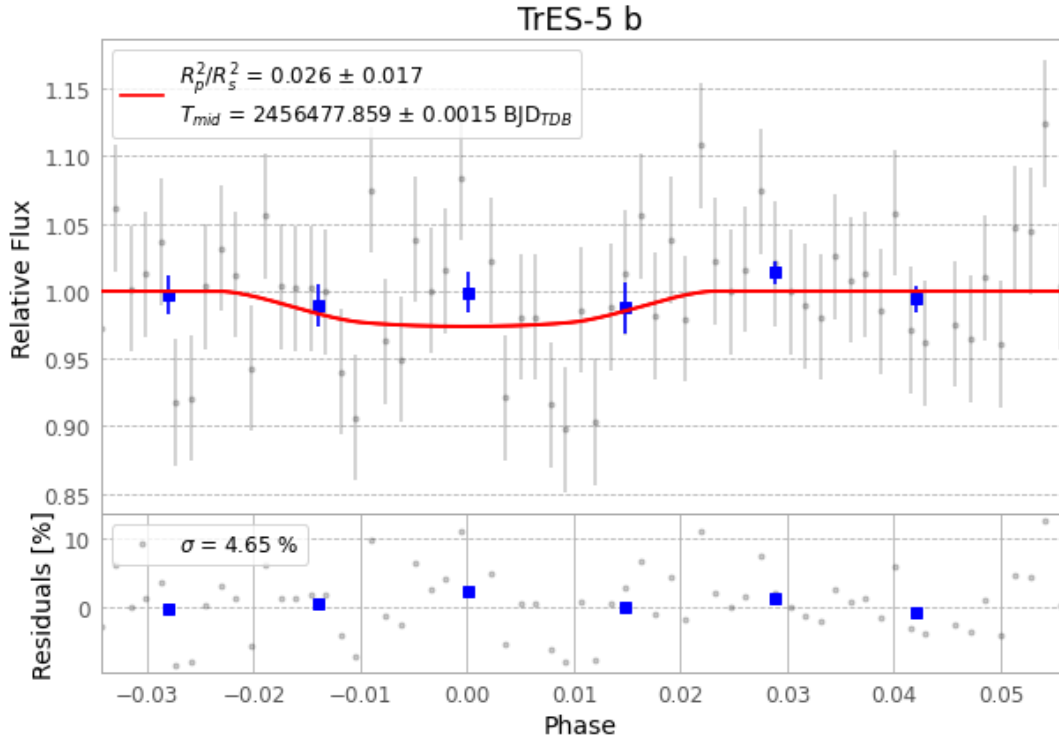


Figure 1 — FinalLightCurve_TrES-5 b_2013-07-04.png (SHA-256 3a06ab173c68bc8b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [467, 227], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1cf726c93197eb28...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

65 FITS frames (43 MB), TRES-5130704072113.FITS → TRES-5130704103313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-130704.log (18c50835f881...), FinalLightCurve_TrES-5 b_2013-07-04.png (3a06ab173c68...), FinalLightCurve_TrES-5 b_2013-07-04.csv (54d6dde45bd0...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 21:05Z.